
Online Examination Report

Date:	21.07.2014
Attempts:	1
Examinee:	Akandwanaho, David
Date of birth:	2/2/1987
Subject:	030-Flight Performance&Planing
Result:	32,49 %
Time used:	00:00:12 (HH:MM:SS)

Attachments

1. List of result by question
2. List of result by topic
3. Detailed question result report

Attachment 1: List of result by question

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
1	384	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
2	379	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	1,00	1,00
3	381	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	1,00	1,00
4	361	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
5	374	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
6	388	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	1,00	1,00
7	315	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
8	375	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	1,00	1,00
9	380	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	1,00	1,00
10	382	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
11	420	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
12	391	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
13	376	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	1,00	1,00
14	183	31.2	LOADING	1,00	1,00
15	257	31.3.1	Definition of centre of gravity	0,00	1,00
16	410	32.	PERFORMANCE - AEROPLANES	0,00	1,00
17	316	32.	PERFORMANCE - AEROPLANES	1,00	1,00
18	61	32.2.5.1	Take-off	0,00	1,00
19	231	32.2.5.4	Landing	0,00	1,00
20	152	32.2.5.4	Landing	0,00	1,00
21	236	32.3.3.1	Take-off	0,00	1,00
22	132	32.3.3.1	Take-off	1,00	1,00
23	179	32.3.3.1	Take-off	1,00	1,00
24	123	32.4.1	Take - off	0,00	1,00
25	95	32.4.1.1	Definitions of terms used	0,00	1,00
26	163	32.4.1.7	Take-off climb	0,00	1,00
27	51	32.4.3	Cruise	0,00	1,00
28	213	32.4.3	Cruise	0,00	1,00

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
29	129	32.4.3	Cruise	0,00	1,00
30	24	32.4.3.5	Influence of variables on cruise performance	0,00	1,00
31	18	33.	FLIGHT PLANNING AND FLIGHT MONITORING	1,00	1,00
32	418	33.	FLIGHT PLANNING AND FLIGHT MONITORING	0,00	1,00
33	92	33.	FLIGHT PLANNING AND FLIGHT MONITORING	0,00	1,00
34	414	33.	FLIGHT PLANNING AND FLIGHT MONITORING	1,00	1,00
35	421	33.	FLIGHT PLANNING AND FLIGHT MONITORING	1,00	1,00
36	148	33.1.1.1	Routes, airfields, heights and altitudes from VFR charts	0,00	1,00
37	2	33.2.1	IFR Navigation plan	0,00	1,00
38	79	33.3.1	General	0,00	1,00
39	224	33.4.2.1	Extraction and analysis of relevant data from meteorological documents	0,00	1,00
40	30	33.5.1.1	Format of Flight Plan	0,00	1,00
Total:32%				13,00	40,00

Attachment 2: List of result by topic

Licence

Date: 18.07.2014

Examinee: Akandwanaho, David

Location: Uganda Civil Aviation Authority

Your Result: 13,00 from 40,00 Points (32,50 %)

Topics	Amount of Questions	Points achieved	Maximum Points	Percent
31. MASS AND BALANCE - AEROPLANES/HELICOPTERS	15	7,00	15,00	46,67
31.2. LOADING	1	1,00	1,00	100,00
31.3. FUNDAMENTALS OF CG CALCULATIONS	1	0,00	1,00	0,00
32. PERFORMANCE - AEROPLANES	15	3,00	15,00	20,00
32.2. PERFORMANCE CLASS B - SINGLE-ENGINE AEROPLANES	3	0,00	3,00	0,00
32.3. PERFORMANCE CLASS B - MULTI-ENGINE AEROPLANES	3	2,00	3,00	66,67
32.4. PERFORMANCE CLASS A - AEROPLANES CERTIFICATED UNDER CS 25 ONLY	7	0,00	7,00	0,00
33. FLIGHT PLANNING AND FLIGHT MONITORING	10	3,00	10,00	30,00
33.1. FLIGHT PLANNING FOR VFR FLIGHTS	1	0,00	1,00	0,00
33.2. FLIGHT PLANNING FOR IFR FLIGHTS	1	0,00	1,00	0,00
33.3. FUEL PLANNING	1	0,00	1,00	0,00
33.4. PRE-FLIGHT PREPARATION	1	0,00	1,00	0,00
33.5. ICAO FLIGHT PLAN (ATS Flight Plan)	1	0,00	1,00	0,00
Total	40	13,00	40,00	32,50

1

An aircraft is loaded with the CG at the aft limit. What are the performance characteristics compared with the CG at the forward limit?

- ☐ The aft CG provides the lowest stalling speed, the highest cruising speed, and least stability.
- ☒ Cruising speed is lower because of more induced drag created by the elevator or stabilizer being required to provide more lift with an aft CG.
- ☐ The aft CG provides the highest stall speed and cruising speed.

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2

How much weight must be shifted from Station 150.0 to Station 30.0 to move the CG to exactly the aft CG limit?

Total weight 7,500 lb

CG location Station 80.5

☐ 65.8 pounds.

☒ 62.5 pounds.

☐ 68.9 pounds.

3

Could 100 pounds of weight be shifted from Station 30.0 to Station 120.0 without exceeding the aft CG limit?

Total weight 4,750 lb

CG location Station 115.8

☒ Yes; the new CG would be located at Station 117.69.

☐ No; the new CG would be located at Station 118.15.

☐ Yes; the CG would remain at Station 115.8.

4

What is the maximum allowable weight that may be carried on a pallet which has the dimensions of 24.6 x 68.7 inches?
Floor load limit - 85 lb/sq ft
Pallet weight - 44 lb
Tiedown devices - 29 lb

☒ 953.6 pounds.

☐ 968.6 pounds.

☐ 924.5 pounds.

5

How much weight could be added at Station 120 without exceeding the aft CG limit?

Aircraft weight 9,500 lb
CG location Station 90.0
Aft CG limit Station 90.5

☒ 110.5 pounds.

☐ 161.0 pounds.

☐ 61.0 pounds.

6

What is characteristic of the indicated airspeed if the CG is at the most forward allowable position and constant power and altitude are maintained?

- ☒ Indicated airspeed will be less than it would be with the CG in the most rearward allowable position.
- ☐ There is no relationship between CG location and indicated airspeed.
- ☐ Indicated airspeed will be greater than it would be with the CG in the most rearward allowable position.

7

The term 'weigh-off' as used in ballooning means to determine the

- ☒ standard weight and balance of the balloon.
- ☐ amount of fuel required for ascent to a preselected altitude.
- ☐ static equilibrium of the balloon as loaded for flight.

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8

What is the maximum weight that could be added at Station 150.0 without exceeding the aft CG limit?

Aircraft weight 5,000 lb
CG location Station 80.0

☐ 70.0 pounds.

☒ 35.9 pounds.

☐ 69.5 pounds.

9

Could 100 pounds of weight be shifted from Station 130.0 to Station 30.0 without exceeding the forward CG limit?

Total weight 2,800 lb
CG location Station 120.0

- ☐ No; the new CG would be located at Station 116.89.
- ☐ Yes; the new CG would be located at Station 117.89.
- ☒ No; the new CG would be located at Station 116.42.

10

The center of gravity of an aircraft can be determined by

- ☐ dividing total arm by total moments.
- ☒ multiplying total arm by total weight.
- ☐ dividing total moment by total weight.

11

The center of gravity tube is

☒ lengthened for heavier pilots.

☐ lengthened for lighter pilots.

☐ shortened for lighter pilots.

12

As the CG location is changed, recovery from a stall becomes progressively

☒ **more difficult as the CG moves either forward or rearward.**

☐ **more difficult as the CG moves rearward.**

☐ **less difficult as the CG moves rearward.**

13

What is the location of the CG if 90 pounds are removed from Station 140?

Aircraft weight 6,230 lb

CG location Station 79

☐ 77.9.

☒ 78.1.

☐ 79.9.

14

In flight the wing of an aircraft containing fuel is subjected to vertical loads that produce a bending moment which is:

- ☐ lowest at the wing root
- ☐ equal to the zero -fuel weight multiplied by the span
- ☐ equal to half the weight of the aircraft multiplied by the semi span
- ☒ highest at the wing root

15

When an aircraft is stationary on the ground, its total weight will act vertically

- ☒ *through its centre of pressure.*
- ☐ *through a point defined as the datum point.*
- ☐ *through its centre of gravity.*
- ☐ *through the main wheels of its undercarriage assembly.*

16

At what minimum speed will dynamic hydroplaning begin if a tire has an air pressure of 70 PSI?

☐ 85 knots.

☐ 75 knots.

☒ 80 knots.

17

Excessively high engine temperatures, either in the air or on the ground, will

- ☒ **cause loss of power, excessive oil consumption, and possible permanent internal engine damage.**
- ☐ **result in damage to heat-conducting hoses and warping of cylinder cooling fans.**
- ☐ **increase fuel consumption and may increase power due to the increased heat.**

18

What effect has a downhill slope on the take-off speeds?The slope

- ☐ has no effect on the take-off speed V_1 .
- ☐ decreases the TAS for take-off.
- ☒ increases the IAS for take-off.
- ☐ decreases the take-off speed V_1 .

19

To minimize the risk of hydroplaning during landing the pilot should:

- ☐ postpone the landing until the risk of hydroplaning no longer exists.
- ☐ make a "positive" landing and apply maximum reverse thrust and brakes as quickly as possible.
- ☐ use normal landing-, braking- and reverse technique.
- ☒ use maximum reverse thrust, and should start braking below the hydroplaning speed.

20

Which of the following factors will lead to an increase of ground distance during a glide, while maintaining the appropriate minimum glide angle speed?

☒ Headwind.

☐ Increase of aircraft mass.

☐ Decrease of aircraft mass.

☐ Tailwind.

21

Provided all other parameters stay constant. Which of the following alternatives will decrease the take-off ground run?

- ☒ Increased outside air temperature, decreased pressure altitude, decreased flap setting.
- ☐ Decreased take-off mass, increased pressure altitude, increased temperature.
- ☐ Increased pressure altitude, increased outside air temperature, increased take-off mass.
- ☐ Decreased take-off mass, increased density, increased flap setting.

22

On a dry runway the accelerate stop distance is increased

☒ by a lower take-off mass because the aeroplane accelerates faster to V1.

☐ by low outside air temperature.

☐ by headwind.

☐ by uphill slope.

23

The decision speed at take-off (V1) is the calibrated airspeed:

- ☐ below which the take-off must be continued.
- ☒ below which take-off must be rejected if an engine failure is recognized, above which take-off must be continued.
- ☐ at which the failure of the critical engine is expected to occur.
- ☐ at which the take-off must be rejected.

24

What will be the effect on an aircraft's performance if aerodrome pressure altitude is decreased?

- ☐ It will increase the take-off distance required.
- ☐ It will increase the accelerate stop distance.
- ☐ It will decrease the take-off distance required.
- ☒ It will increase the take-off ground run.

25

The decision speed at take-off (V1) is the calibrated airspeed:

- ☐ *below which the take-off must be continued.*
- ☒ *at which the take-off must be rejected.*
- ☐ *below which take-off must be rejected if an engine failure is recognized, above which take-off must be continued.*
- ☐ *at which the failure of the critical engine is expected to occur.*

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26

The maximum rate of climb that can be maintained at the absolute ceiling is:

☐ 125 ft/min

☐ 100 ft/min

☒ 500 ft/min

☐ 0 ft/min

27

Which statement with respect to the step climb is correct?

- ☒ A step climb is executed because ATC desires a higher altitude.
- ☐ Executing a desired step climb at high altitude can be limited by buffet onset at g-loads larger than 1.
- ☐ A step climb must be executed immediately after the aeroplane has exceeded the optimum altitude.
- ☐ A step climb is executed in principle when, just after leveling off, the 1.3g altitude is reached.

28

A twin jet aeroplane is in cruise, with one engine inoperative, and has to overfly a high terrain area. In order to allow the greatest clearance height, the appropriate airspeed must be the airspeed

- ☐ of greatest lift-to-drag ratio.
- ☒ for long-range cruise.
- ☐ giving the highest C_d/C_l ratio.
- ☐ giving the lowest C_l/C_d ratio.

29

A constant headwind component

- ☒ increases the best rate of climb.
- ☐ increases the maximum endurance.
- ☐ decreases the angle of climb.
- ☐ increases the angle of flight path during climb.

30

The induced drag of an aeroplane

- ☐ is independent of the airspeed.
- ☐ decreases with increasing airspeed.
- ☐ decreases with increasing gross weight.
- ☒ increases with increasing airspeed.

31

Which of the following statements regarding filing a flight plan is correct?

- ☐ A flying college can file repetitive flight plan for VFR flights.
- ☐ A flight plan should be filed when a national FIR boundary will be crossed.
- ☐ Any flight plan should be filed at least 10 minutes before departure.
- ☒ A flight plan should be filled atleast one hour before departure

32

If an aircraft has a transponder, encoding altimeter, and DME, the proper equipment suffix to be entered on a flight plan is

☒ R.

☐ A.

☐ U.

33

From which of the following would you expect to find details of the Search and Rescue organisation and procedures (SAR) ?

- ☐ ATCC broadcasts
- ☒ SIGMET
- ☐ NOTAM
- ☐ AIP (Aeronautical Information Publication)

34

GIVEN:

Usable fuel at takeoff 40 gal
Fuel consumption rate 12.2 gal/hr
Constant groundspeed 120 kts
Flight time since takeoff 1 hr 30 min

☒ 176 NM.

☐ 121 NM.

☐ 215 NM.

35

How is your flight plan closed when your destination airport has IFR conditions and there is no control tower or Flight control Center(FCC) on the field?

- ☒ Upon landing, you must close your flight plan by radio or by telephone to any FCC or ATC facility.
- ☐ You may close your flight plan any time after starting the approach by contacting any FCC or ATC facility.
- ☐ The ARTCC controller will close your flight plan when you report the runway in sight.

36

From which of the following would you expect to find information regarding known short unserviceability of VOR, TACAN, and NDB ?

☒ B AIP (Aeronautical Information Publication)

☐ D ATCC broadcasts

☐ C SIGMET

☐ A NOTAM

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37

In an ATS flight plan, Item 15 (route), a cruising pressure altitude of 32000 feet would be entered as :

☐ C 32000

☐ B S3200

☒ A FL320

☐ D F320

38

The fuel burn off is 200 kg/h with a relative fuel density of 0,8. If the relative density is 0.75 the fuel burn will be:

☐ 213 kg/h

☐ 267 kg/h

☐ 200 kg/h

☒ 188 kg/h

39

A METAR reads : SA1430 35002KT 7000 SKC 21/03 Q1024 =
Which of the following information is contained in this METAR?

☐ *temperature/dewpoint*

☒ *day/month*

☐ *period of validity*

☐ *runway in use*

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If your destination airport has no ICAO indicator, in the appropriate box of your flight plan, you write:

☒ B XXXX

☐ D ZZZZ

☐ C IIII

☐ A AAAA